

Adversarial Attack Resilience in Autonomous Vehicle Perception

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Abstract—Deep learning-based perception systems for autonomous vehicles are highly vulnerable to physically realizable adversarial attacks, yet remain widely deployed in safety-critical applications such as traffic sign recognition. This paper investigates the adversarial resilience of YOLOv8 [3], a state-of-the-art object detection model, against four attack types – digital tape, adversarial patch, graffiti, and illumination – on a 264-class traffic sign dataset. We develop a custom adversarial data generation pipeline and train 10 fine-tuned model variants at two clean-to-attacked training ratios (50/50 and 70/30), evaluating each on both clean and adversarial validation sets using mAP50 as the primary metric. Our results show that the baseline model suffers significant performance degradation under all four attack types, with mAP50 dropping from 0.5069 on clean data to 0.2163 under adversarial conditions. Attack-specific fine-tuning consistently recovers performance, with the best-performing specialized models achieving mAP50 of 0.5549 on their target attack. Generalization offers a practical compromise between specialization and broad resilience. These findings highlight the feasibility of adversarial fine-tuning as a lightweight model security update strategy and motivate further exploration of universal adversarial defence mechanisms for autonomous vehicle perception.

Index Terms—adversarial attacks, autonomous vehicles, YOLOv8, traffic sign recognition, adversarial fine-tuning, object detection, robustness.

I. INTRODUCTION

Autonomous vehicle perception systems rely on deep learning models to interpret the environment in real time, making decisions about speed, navigation, and safety based on the objects they detect. Among these tasks, traffic sign recognition is particularly critical – a misclassified stop sign or speed limit can directly lead to dangerous driving behaviour. The consequences of such failures extend beyond theoretical concern. High-profile incidents involving perception failures in deployed autonomous systems have underscored the real-world cost of overconfident, brittle models. As autonomous

and semi-autonomous vehicles become increasingly prevalent – with traffic sign recognition now standard across at least 13 of the 15 top-selling vehicle brands in North America [6] – the attack surface exposed to malicious actors grows proportionally. Modern object detectors such as YOLOv8 achieve strong performance under standard conditions, but their reliance on statistical patterns in training data makes them susceptible to adversarial perturbations – carefully crafted modifications to inputs that cause systematic misclassifications [1], [4], [7].

Unlike digital adversarial attacks, which manipulate pixel values in software, physical adversarial attacks introduce perturbations into the real world that persist across varying viewpoints, distances, and lighting conditions [2], [5]. This distinction is especially important for autonomous vehicles, where a piece of tape, a patch of paint, or unusual lighting on a traffic sign constitutes a plausible real-world threat requiring no access to the underlying software system. Despite growing awareness of these vulnerabilities [6], practical strategies for hardening deployed models against such attacks remain underexplored, particularly in the context of lightweight, real-time detectors. While the catalogue of known attack methods has expanded rapidly over the past decade, the development of practical defences has not kept pace. Most proposed mitigation strategies, such as adversarial training [8] and input preprocessing, have been evaluated primarily on image classifiers trained on small, well-curated benchmarks – leaving the robustness of real-time multi-class object detectors like YOLOv8 largely uncharacterized under physical threat conditions.

This paper addresses this gap through three concrete contributions. First, we design and implement a modular adversarial data generation pipeline in Python that applies four physically motivated attack types – digital tape, adversarial patch, graffiti, and illumination – directly to bounding box regions within a 264-class YOLO-format traffic sign dataset,

enabling controlled and reproducible adversarial evaluation. Second, we train and evaluate 10 fine-tuned YOLOv8 model variants under two clean-to-attacked training ratios (50/50 and 70/30), systematically comparing attack-specific and mixed adversarial training strategies against a clean-data baseline. Third, we conduct a cross-attack generalization analysis to determine whether a model hardened against one attack type acquires meaningful robustness against others – a question with direct implications for the feasibility of deploying a single updated model in response to newly observed real-world threats.

Our experiments demonstrate that the baseline YOLOv8 model experiences substantial mAP50 degradation under all attack types, confirming the severity of this vulnerability in practice. Fine-tuning on attack-specific data substantially recovers performance, and mixed adversarial training achieves the strongest overall outcome – improving adversarial mAP50 by 37.1% over the baseline while simultaneously exceeding baseline clean-data performance – establishing adversarial fine-tuning as a viable lightweight security strategy. These results have direct implications for how perception models in autonomous systems might be updated in response to newly discovered real-world threats, suggesting that targeted retraining on observed attack distributions represents a practical and computationally accessible hardening mechanism.

The remainder of this paper is organized as follows. Section II reviews related work on adversarial attacks and defences in the context of autonomous vehicle perception. Section III describes our research objectives, methodology, and dataset. Section IV presents and analyzes our experimental results. Section V concludes with a summary of findings, lessons learned, and directions for future work.

II. BACKGROUND & RELATED WORK

Deep learning has fundamentally transformed how autonomous vehicles perceive their environment. Object detection models, including convolutional neural network-based architectures, achieve strong performance on standard benchmarks; however, their accuracy degrades significantly when test conditions deviate from the training distribution [4], [7]. This brittleness is especially dangerous in safety-critical systems, where even a single misclassification such as a missed stop sign can directly cause physical harm.

The existence of adversarial examples was first demonstrated by Szegedy et al. [7], who showed that imperceptible pixel-level perturbations to an input image could cause a well-trained deep neural network to misclassify it with high confidence. Goodfellow et al. [4] extended this finding by introducing the Fast Gradient Sign Method (FGSM), a computationally efficient technique for generating such perturbations by stepping in the direction of the loss gradient. These early works established that adversarial vulnerability is

a fundamental property of gradient-based learning, not merely a consequence of insufficient training data or model capacity.

Adversarial attacks can be broadly categorized along two axes: digital vs. physical, and white-box vs. black-box. Digital attacks manipulate pixel values directly in software, while physical attacks introduce perturbations into the real world – for instance, by printing stickers or altering lighting conditions. White-box attacks assume full knowledge of the model architecture and weights, whereas black-box attacks operate with only query access to the model output [6]. Physical, black-box attacks are the most relevant to autonomous vehicle deployment, as they represent realistic threat vectors that do not require insider access to the target system.

Eykholt et al. [5] were among the first to demonstrate that physical perturbations could reliably fool deep learning classifiers in real-world conditions. Using their Robust Physical Perturbations (RP2) algorithm, they applied carefully positioned black-and-white sticker patterns to physical stop signs, achieving targeted misclassification in 100% of controlled lab settings and 84.8% of real-world driving frames. This work established that adversarial attacks are not merely a theoretical concern – they transfer effectively to physical deployment scenarios under varying viewpoint, distance, and lighting conditions.

Building on this, Kong et al. [2] introduced PhysGAN, a generative adversarial network-based approach to producing adversarial examples specifically designed to remain effective across the physical variations encountered in real driving scenarios, including camera angle changes and illumination shifts. Unlike earlier perturbation methods, PhysGAN models the full distribution of physical transformations during adversarial example generation, producing attacks that are substantially more robust to environmental variance. This highlights the importance of evaluating adversarial robustness not only in digital settings but under realistic physical conditions.

Shah [1] specifically examined adversarial resilience in autonomous driving perception systems, introducing the attack taxonomy most relevant to our work: tape, graffiti, and illumination-based attacks. Tape attacks place strategically positioned occlusions across traffic sign faces (Fig. 1). Graffiti attacks introduce randomized coloured overlays that obscure sign features. Illumination attacks manipulate regional brightness and contrast to impair sign visibility. Each of these attack types is low-cost and easily deployable by a malicious actor in an urban environment, making them highly relevant threat vectors for real-world autonomous driving systems.

More broadly, Pavlitska et al. [6] survey the landscape of adversarial attacks specifically targeting traffic sign recognition systems, covering both digital and physical attack methodologies. Their survey highlights that while attack methods have grown increasingly sophisticated, defensive strategies for traffic sign recognition – particularly in the context of lightweight, real-time detectors – remain comparatively un-

derexplored. This gap directly motivates our work, which evaluates whether adversarial fine-tuning can serve as a practical post-deployment defence mechanism for YOLOv8-based traffic sign detection. Unlike prior work that primarily targets classification models on datasets such as GTSRB, our study operates on a 264-class detection dataset using the YOLOv8 object detection architecture, enabling a more direct evaluation of the risks facing real autonomous vehicle perception pipelines.

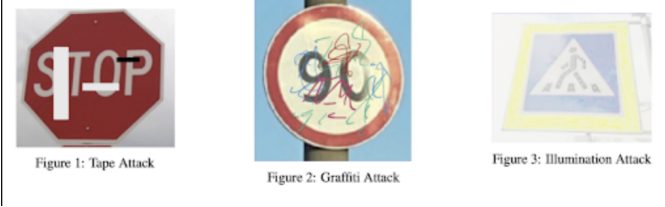


Fig. 1. Examples of physical adversarial attacks on traffic signs: (left) digital tape, (center) graffiti, (right) illumination [1].

III. METHODS

A. Research Objectives

The main goal of this study is to understand the vulnerabilities of YOLOv8 under adversarial threats and to evaluate the effectiveness of adversarial fine-tuning as a defence.

O1 – Measure Baseline Vulnerability: Measure the performance loss of a standard YOLOv8 model when exposed to four adversarial attack types (tape, patch, graffiti, illumination). This establishes a foundation for the research.

O2 – Evaluate Adversarial Fine-tuning: Determine whether fine-tuning a model on a specific attack type improves its mAP50 on both clean and attacked datasets, establishing whether patching model weights is a viable security update strategy.

O3 – Analyze Generalization: Determine whether specialized training for one attack type helps defend against other attack types, contributing to understanding whether a single defence can address multiple simultaneous threats.

B. Research Methodology

The research is conducted as an empirical study, with a development pipeline created to generate adversarial datasets and evaluate model performance.

1) *Dataset and Technical Approach:* A road sign detection dataset from Kaggle, built specifically for YOLOv8, is used as the primary dataset. To simulate real-world threats, a custom adversarial attack generator was developed in Python to apply attacks to the bounding box regions of traffic signs in each image, inspired by Shah [1]. Four attack types are implemented:

- **Digital tape:** horizontal occlusion representing tape applied to the traffic sign.
- **Adversarial patch:** a patch covering 50% of the sign with random high-frequency noise.
- **Graffiti:** randomized, coloured shape overlays applied to the sign region.
- **Illumination:** brightness shifts simulating extreme lighting conditions.

The dataset used in this study is a 264-class traffic sign detection dataset sourced from Kaggle, formatted natively for YOLOv8 with per-image label files containing normalized bounding box coordinates and class indices. The dataset is pre-split into training and validation sets. All adversarial perturbations are applied exclusively to the validation set, preserving the integrity of the training distribution for baseline model evaluation. The attack generator, implemented in Python using OpenCV, operates directly on bounding box regions identified in each image's label file, ensuring that perturbations are anatomically targeted to the sign face rather than applied globally to the image.

The YOLOv8n (nano) variant is used throughout all experiments, selected for its lightweight architecture suitable for edge deployment in real autonomous vehicle hardware. The baseline model is trained from the official pretrained YOLOv8n.pt weights for 100 epochs at image resolution 640×640 , with a batch size of 16, mosaic augmentation ($p = 1.0$), and mixup augmentation ($p = 0.1$). All training is performed on a single NVIDIA T4 GPU via Google Colab.



Fig. 2. Example of a physical adversarial patch attack on a stop sign, where printed perturbations are applied directly to the sign surface [9].



Fig. 3. Examples of adversarially attacked traffic signs and their resulting misclassifications by a deep learning model [6].

2) *Implementation Strategy*: Models are trained and fine-tuned on attack datasets using two clean-to-attack ratios: 50/50 and 70/30. This is done to identify the optimal balance in an adversarial training set for achieving the highest mAP50 scores under both clean and attacked conditions. Every model is cross-validated on each attack type, as well as a generalized adversarial dataset using a modulo indexing strategy to ensure even distribution across all four attack types. This setup measures specialized versus general resilience for each model.

Sequential fine-tuning is used as the training strategy: each successive model is initialized from the weights of the previous model rather than from scratch. The training order is Baseline $\rightarrow$ Tape $\rightarrow$ Patch $\rightarrow$ Illumination $\rightarrow$ Graffiti $\rightarrow$ Mixed. For each attack-specific model, the adversarial training set is constructed by replacing the corresponding proportion of clean training images with adversarially perturbed versions at the specified clean-to-attack ratio. The mixed adversarial training set is constructed using a modulo indexing strategy that cycles evenly across all four attack types, ensuring balanced representation across the combined adversarial distribution. Every model variant is evaluated on five validation sets: clean, tape-attacked, patch-attacked, illumination-attacked, and graffiti-attacked.

C. Performance Analysis Methods

Evaluation Metric: The primary metric is mAP50 (mean average precision at 0.5 intersection over union). Precision is also reported to assess false detections caused by adversarial noise.

Model Analysis: Performance is compared against a baseline model trained only on clean data to quantify attack defence improvement. The effect of sequential fine-tuning is also analyzed to determine whether repeated dataset exposure improves clean mAP over time.

D. Threats to Validity

Internal validity: Sequential fine-tuning (Baseline $\rightarrow$ Tape $\rightarrow$ Patch $\rightarrow$ Illumination $\rightarrow$ Graffiti $\rightarrow$ Mixed) introduces a bias where mAP50 may cumulatively increase with each new model due to additional training epochs rather than genuine architectural improvement.

External validity: Although the dataset is relatively robust, it may contain too few images per class, which could cause the model to struggle on underrepresented categories. Furthermore, the images may not fully capture real-world conditions such as adverse weather.

Metric validity: mAP is a standard metric, but in an autonomous vehicle context, a single missed stop sign carries safety implications that a small aggregate drop in mAP may not adequately reflect.

IV. RESULTS

Table I summarizes mAP50, mAP50-95, and Precision for all 10 model variants evaluated on both clean and adversarial validation sets. Fig. 5 provides a visual comparison of mAP50 scores across training strategies. The results reveal four key patterns: (1) significant baseline vulnerability to all attack types; (2) consistent recovery through attack-specific fine-tuning; (3) preserved or improved clean-data performance after adversarial training; and (4) superior generalization from mixed adversarial training.

TABLE I
PERFORMANCE COMPARISON OF YOLOV8 MODELS UNDER CLEAN AND ADVERSARIAL CONDITIONS

Model	Dataset	mAP50	mAP50-95	Precision
Baseline	Clean	0.5069	0.4383	0.5991
Baseline	Attack	0.2163	0.1597	0.5704
50/50 Tape	Clean	0.5781	0.4887	0.5955
50/50 Tape	Attack	0.3486	0.2886	0.6588
70/30 Tape	Clean	0.5845	0.4960	0.5859
70/30 Tape	Attack	0.3048	0.2363	0.5819
50/50 Patch	Clean	0.5668	0.5054	0.5147
50/50 Patch	Attack	0.3890	0.3048	0.4372
70/30 Patch	Clean	0.6160	0.5315	0.5223
70/30 Patch	Attack	0.3723	0.2734	0.5417
50/50 Illumination	Clean	0.6303	0.5444	0.5436
50/50 Illumination	Attack	0.3279	0.2631	0.4613
70/30 Illumination	Clean	0.6233	0.5320	0.6011
70/30 Illumination	Attack	0.3351	0.2619	0.5236
70/30 Graffiti	Clean	0.6657	0.5781	0.4930
70/30 Graffiti	Attack	0.4413	0.3413	0.5787
50/50 Mixed	Clean	0.6952	0.6190	0.5705
50/50 Mixed	Attack	0.5549	0.4940	0.5446
70/30 Mixed	Clean	0.6544	0.5759	0.5788
70/30 Mixed	Attack	0.4431	0.3940	0.4752

A. Baseline Performance

The baseline model, trained exclusively on clean data, achieves an mAP50 of 0.5069 on the clean validation set, confirming that the model learns effective traffic sign detection features under standard conditions. However, performance degrades substantially under adversarial conditions, with mAP50 falling to 0.2163 – a relative drop of approximately 57.3% – demonstrating that the baseline model is highly vulnerable to physically realizable perturbations. This confirms O1: standard YOLOv8 models trained only on clean data are not robust to the attack types considered in this study. The degradation is consistent with the broader findings of Eykholt et al. [5] and Goodfellow et al. [4], who established that neural networks trained on unperturbed data fail to learn invariance to adversarial perturbations.

B. Effect of Adversarial Fine-tuning

Attack-specific fine-tuning consistently recovers detection performance on the corresponding adversarial validation set. The 50/50 Tape model improves adversarial mAP50 from 0.2163 to 0.3486, and the 70/30 Patch model achieves the highest clean-data performance of any model at 0.6160, suggesting that adversarial training with a higher proportion of clean images better preserves feature representations learned from unperturbed data. These results support O2: fine-tuning a model on a specific attack type does improve its mAP on attacked inputs, and this improvement does not require sacrificing clean-data performance.

Notably, the relationship between the clean-to-attack ratio and performance is not uniform across attack types. For tape and patch, the 70/30 ratio outperforms 50/50 on clean data, whereas the difference narrows for illumination and graffiti. This suggests that tape and patch attacks introduce stronger distributional shifts that require more conservative training ratios to prevent over-adaptation to the adversarial distribution.

C. Performance on Clean Data

Clean data performance is preserved, and often improved, by adversarial fine-tuning. The 70/30 patch model achieves mAP50 of 0.6160 on clean data, compared to the baseline’s 0.5069, suggesting that adversarial training also improves general feature learning.

D. Generalization Across Attack Types

A key finding relevant to O3 is the degree to which models trained on one attack type generalize to others. Models fine-tuned on a single attack type show partial but inconsistent cross-attack generalization. The 70/30 Graffiti model, which achieves 0.4413 mAP50 on adversarial inputs, also shows above-baseline performance on tape and illumination validation sets, suggesting that graffiti forces the model to learn more broadly robust low-level feature representations. In contrast,

tape-trained models generalize less effectively, likely because tape attacks represent a simpler, more structured occlusion pattern that the model learns to specifically compensate for without developing broader invariance. These findings indicate that cross-attack generalization is limited when using single-attack training, partially supporting O3.

E. Mixed Adversarial Training

The 50/50 Mixed model achieves the strongest overall result, reaching mAP50 of 0.5549 on adversarial inputs while maintaining 0.6952 on clean data – a 37.1% improvement over the adversarial baseline while simultaneously exceeding baseline clean-data performance. This is the most practically significant finding of this study: a single model trained on a balanced mixture of all four attack types outperforms all single-attack-specialized models on the combined adversarial validation set, without sacrificing clean-data accuracy. This result is consistent with the adversarial training theory established by Madry et al. [8], which demonstrates that models trained against a diverse adversarial distribution develop more generalizable internal representations.

F. Precision Analysis

Precision metrics across models reveal an interesting secondary finding: several adversarially trained models exhibit higher precision than the baseline on clean data. This suggests that adversarial training acts as a form of regularization, reducing false positive detections that may be caused by the model latching onto superficial texture features in the training set. This effect is most pronounced in the 70/30 Patch and 50/50 Mixed models, both of which achieve precision values above the baseline on clean validation.

G. Training Behaviour of the 70/30 Mixed Model

The training behaviour of the 70/30 mixed model shows a clear and stable learning trend over time. Across epochs, all loss components (box loss, classification loss, and distribution focal loss) consistently decrease, indicating that the model is effectively learning meaningful features. At the same time, performance metrics such as mAP50, mAP50–95, precision, and recall steadily improve, with only small fluctuations early in training. The validation curves follow a similar pattern to the training curves, which suggests the model is not overfitting and is generalizing well to unseen data. The steady increase in mAP50, reaching values above 0.64, further supports the effectiveness of mixed adversarial training. Overall, these results reinforce that combining clean and adversarial data leads to both stable convergence and improved robustness, without hurting performance on clean inputs.

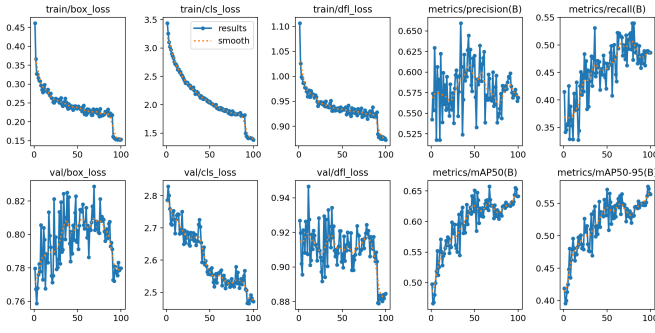


Fig. 4. Training and validation curves for the 70/30 mixed model over 100 epochs, showing loss components and performance metrics.

H. Overall Findings

Adversarial fine-tuning consistently improves resilience without degrading standard performance, and frequently enhances it. Mixed adversarial training stands out as the most effective strategy, supporting both precision and durability across attack types, making it well-suited for real-world autonomous vehicle perception.

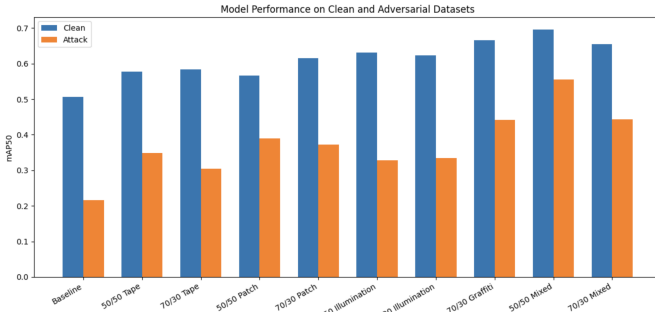


Fig. 5. Comparison of mAP50 performance across training strategies on clean and adversarial datasets.

V. CONCLUSIONS & FUTURE WORK

This study investigated the adversarial vulnerability of YOLOv8 traffic sign detection and evaluated adversarial fine-tuning as a practical defence strategy. Addressing O1, we confirmed that the baseline model is highly susceptible to all four physically realizable attack types – tape, patch, graffiti, and illumination – with mAP50 dropping from 0.5069 on clean data to 0.2163 under adversarial conditions, a relative degradation of 57.3%. Addressing O2, attack-specific fine-tuning consistently recovered detection performance without sacrificing clean-data accuracy, and in several cases improved upon the baseline on clean inputs, supporting the feasibility of adversarial fine-tuning as a lightweight post-deployment security update strategy. Addressing O3, cross-attack generalization was observed but inconsistent, with graffiti-trained models demonstrating the broadest transferable robustness and tape-trained models the most attack-specific. Most significantly, the 50/50 Mixed model achieved the strongest overall

performance across both clean and adversarial conditions, demonstrating that mixed adversarial training is the most practical single-model defence against an unknown adversarial threat landscape.

Several important limitations must be acknowledged. The sequential fine-tuning strategy introduces cumulative epoch exposure as a confound: later models have effectively seen the dataset more times than earlier ones, meaning that some clean-data performance gains may reflect additional training time rather than improved architecture or defence. Future work should control for this by training all model variants from scratch using identical epoch budgets. Additionally, the dataset, while broad in class count at 264 classes, contains relatively few images per class, which may limit the model’s ability to generalize within individual sign categories. Expanding to larger datasets such as GTSRB or Mapillary Traffic Sign Dataset would strengthen the external validity of these findings. Finally, the adversarial attacks used in this study are implemented digitally rather than physically – future work should evaluate model robustness against perturbations applied to physical signs photographed under real driving conditions, accounting for factors such as weather, motion blur, viewing angle, and distance. Extending the defence framework to incorporate detection-based input screening [8] or ensemble approaches may further improve robustness without the computational overhead of retraining.

REFERENCES

- [1] A. Shah, “Adversary ML resilience in autonomous driving through human-centered perception mechanisms,” arXiv preprint arXiv:2305.05392, 2023.
- [2] Z. Kong, J. Guo, A. Li, and C. Liu, “PhysGAN: Generating physical-world-resilient adversarial examples for autonomous driving,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2020, pp. 14254–14263.
- [3] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” version 8.0.0, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [4] I. J. Goodfellow, J. Shlens, and C. Szegedy, “Explaining and harnessing adversarial examples,” in *Proc. Int. Conf. Learning Representations (ICLR)*, San Diego, CA, USA, 2015.
- [5] K. Eykholt, I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, and D. Song, “Robust physical-world attacks on deep learning visual classification,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, Salt Lake City, UT, USA, 2018, pp. 1625–1634.
- [6] S. Pavlitska, N. Lambing, and J. M. Zöllner, “Adversarial attacks on traffic sign recognition: A survey,” in *Proc. 3rd Int. Conf. Electrical, Computer, Communications and Mechatronics Engineering (ICEC-CME)*, Tenerife, Spain, Jul. 2023, pp. 1–6.
- [7] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, “Intriguing properties of neural networks,” in *Proc. Int. Conf. Learning Representations (ICLR)*, Banff, Canada, 2014.
- [8] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, “Towards deep learning models resistant to adversarial attacks,” in *Proc. Int. Conf. Learning Representations (ICLR)*, Vancouver, Canada, 2018.
- [9] T. B. Brown, D. Mané, A. Roy, M. Abadi, and J. Gilmer, “Adversarial patch,” in *Proc. NIPS Workshop on Machine Deception*, Long Beach, CA, USA, 2017.